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CAPSULE SUMMARY (50/50)

- Surgical margin size in Merkel cell carcinoma is controversial, however, we found that if adjuvant radiotherapy (aRT) is given, patients with narrow or even microscopically-positive margins have excellent local disease control.
- If aRT is planned, narrower surgical margins are adequate, reduce morbidity, and minimize delay in initiating radiotherapy.

Narrow excision margins are appropriate for Merkel cell carcinoma when combined with adjuvant radiation: Analysis of 188 cases of localized disease and proposed management algorithm

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ABSTRACT:

Background: Merkel cell carcinoma (MCC) management typically includes surgery with or without adjuvant radiation therapy (aRT). Major challenges include determining surgical margin size and whether aRT is indicated.

Objective: To assess the association of aRT, surgical margin size, and MCC local recurrence.

Methods: Analysis of 188 MCC cases presenting without clinical nodal involvement.

Results: aRT-treated patients tended to have higher-risk tumors (larger diameter, positive microscopic margins, immunosuppression) yet had fewer local recurrences (LR) than patients treated with surgery only (1% vs. 15%; $p=0.001$). For patients who underwent surgery alone, 7 of 35 (20%) treated with narrow margins (defined as ≤ 1.0 cm) developed LR, whereas 0 of 13 patients treated with surgical margins >1.0 cm developed LR ($p=0.049$). For aRT-treated patients, local control was excellent regardless of surgical margin size; only 1% recurred in each group (1 of 70 with narrow margins ≤ 1 cm and 1 of 70 with margins >1 cm, $p=0.56$).

Limitations: This was a retrospective study.

Conclusions: Among patients treated with aRT, local control was superb even if significant risk factors were present and margins were narrow. We propose an algorithm for managing primary MCC that integrates risk factors and optimizes local control while minimizing morbidity.

CAPSULE SUMMARY (50/50)

- Surgical margin size in Merkel cell carcinoma is controversial, however, we found that if adjuvant radiotherapy (aRT) is given, patients with narrow or even microscopically-positive margins have excellent local disease control.
- If aRT is planned, narrower surgical margins are adequate, reduce morbidity, and minimize delay in initiating radiotherapy.

INTRODUCTION

Merkel cell carcinoma (MCC) is a rare, aggressive cutaneous neuroendocrine malignancy. In 2015, there were approximately 2,500 cases/year; this is expected to increase to 3,300 by 2025¹. Approximately 65% of MCC patients present with localized disease based on national cancer registry data². Published local recurrence (LR) rates are highly variable because historical cohorts are retrospective, heterogeneous, and often combine LRs with in-transit and regional recurrences. Nevertheless, LRs arise in 4% to 30% of patients³⁻⁷.

Current guidelines for the treatment of primary cutaneous MCC include wide local excision (WLE), sentinel lymph node biopsy (SLNB) for pathologic staging, and consideration of adjuvant radiation therapy (aRT). The National Comprehensive Cancer Network (NCCN) currently recommends WLE with “1-2 cm margins with intent to obtain negative margins when possible”⁸. However, there is no consensus on the appropriate surgical margin size.

Prior studies on the relationship between surgical margin size and LR have yielded seemingly conflicting results. An early study of 70 patients found a trend towards fewer recurrences in patients treated with surgical margins >3.0 cm⁹. However, more recent studies found no difference in LR rates when comparing ≤ 1.0 cm versus >1.0 cm margins^{3, 5} or with >2.0 cm margins compared to ≤ 2.0 cm margins⁴. A relevant limitation of some prior studies is that they do not separately consider surgical margins in the absence and presence of aRT. This is important because numerous studies indicate aRT markedly decreases the rate of LR¹⁰⁻¹³.

An exception is a study of 179 Canadian MCC patients by Harrington et al which separated patients based on aRT status. They found that patients treated with aRT had a low LR rate (6.8%; n=132) regardless of margin size¹⁴. A separate study of low risk MCC patients (primary <2 cm) who did not receive aRT also found a low ($<1\%$) recurrence rate regardless of margin size¹⁵.

We sought to determine the relationship between surgical margin size, aRT, and local disease control in a large, single-center cohort of patients who presented with a primary cutaneous MCC tumor.

METHODS

We performed a retrospective analysis using a Seattle-based repository of >1,400 MCC patients that has been enrolling subjects since 2003. We included MCC patients with local and SLNB detected microscopic nodal disease and excluded patients who had advanced disease including muscle/bone invasion, clinically/radiologically evident nodal disease, and metastatic disease. Patients were excluded if surgical margin, radiation therapy, and follow-up data were unavailable or if patients had occult disease. Patients enrolled >180 days from diagnosis were excluded to eliminate ascertainment bias associated with late referral to a tertiary center that could diminish how accurately the cohort represents the natural history of the disease.

Clinical characteristics collected included age, sex, stage, site of primary (head/neck versus trunk/extremities), size of primary tumor (≤ 1 cm, 1-2 cm, or >2 cm), and presence of immunosuppression (HIV, organ transplant, hematologic malignancy, or chronic use of T cell immunosuppressive medication¹⁶). Treatment characteristics were obtained through chart review. Patients were separated based on whether they received aRT to the primary tumor bed after undergoing re-excision. These groups were further separated by surgical margin size of the primary tumor as noted in operative reports (≤ 1 cm versus >1 cm; **Figure 1**). Patients who underwent Mohs micrographic surgery were excluded.

Recurrences were categorized into four groups: local, in-transit, regional, and metastatic. *Local* was a recurrence arising within or adjacent to the primary excision scar and within 2 cm of the primary tumor site; *in-transit* as cutaneous/subcutaneous lesion not involving regional lymph

nodes and arising >2 cm from the primary scar; *regional* as arising in the draining lymph node basin; and *metastatic* as occurring beyond the draining lymph node basin.

Statistical analyses were performed using the Stata software version 14.0 (StataCorp, College Station, TX). Fisher's exact or Wilcoxon rank-sum tests were used to compare clinical and tumor characteristics. LR outcomes were measured by the permutation test, with non-local MCC recurrences and death as competing risks.

A p-value of <0.05 was considered to be statistically significant. Competing risk regression curves were used to represent local recurrence-free survival. Patients were grouped based on their primary tumor site, aRT status, and margin size. Local recurrence-free survival was defined as the length of time from the surgery date to the date of MCC LR, last follow-up, or death. All studies were performed in accordance with Helsinki principles and approved by the Fred Hutchinson Cancer Research Center Institutional Review Board (IRB #6585).

RESULTS

Patient and Tumor Characteristics

One hundred eighty-eight patients were identified from the repository, of which 140 were treated with surgery and aRT while 48 were treated with surgery alone (**Figure 1**). The majority of patients were male, ≥65 years old, and immunocompetent. Of the patients who underwent successful SLNB, 24% had pathological evidence of tumor invasion into the lymph nodes (**Table 1**). Median follow up for the entire cohort was 4 years (range 42 days–12 years).

The majority of patients had surgical margins of either 1 cm (81 patients) or 2 cm (59 patients) with the remaining 48 patients having other margin sizes. For several known risk factors, patients that received aRT had higher risk tumors as compared to patients who received surgery alone. Specifically, tumor size was larger for aRT-treated patients (29% of tumors ≥2 cm vs 8% for patients treated with surgery alone; $p<0.001$). In addition, aRT-treated patients had higher stage (IIIA) tumors (30% compared to 6.2% in the surgery only group; $p=0.001$).

However, they were less likely to have their primary tumors on the head/neck (26% versus 63% in the surgery-only cohort; $p<0.001$). Patients who received aRT with surgical margins ≤ 1 cm were more likely to have a primary tumor on the head and neck ($p<0.003$) and smaller primary tumor size ($p<0.020$). For the patients that received surgery only to their primary tumor, the only significant difference in patient characteristics between margin groups was that patients with tumors on the head and neck were more likely to receive smaller surgical margins ($p=0.049$; **Table 1**).

Local Recurrences

Among the 188 patients, there were 9 LR (s) (**Table 2**). After adjusting for margin size and aRT status, more LR (s) occurred on the head/neck versus trunk/extremities ($p=0.013$) (**Table 2**). Seven of the 9 patients with a LR were salvaged with surgery, radiation, and in one case, immunotherapy.

In the 140 patients treated with surgery and aRT, 2 LR (s) occurred; for the smaller margin group, there was one LR on the head/neck and for the larger surgical margin group there was one LR on the lower limb. There was no statistically significant difference in local recurrence-free survival between the surgical margin groups ($p=0.56$) (**Figure 3**).

In the 48 patients treated with surgery only, 7 LR (s) occurred, all of which were on the head/neck. There were 7 recurrences in the smaller margin group and none in the larger margin group. There was a significant difference in local recurrence-free survival between the surgical margin size groups ($p=0.049$) (**Figure 2**).

MCC-Specific Survival

In addition to the 9 LR (s), there were 8 in-transit, 15 nodal, and 22 distant recurrences. Thirty-seven patients died during follow up of which 21 were caused by MCC. While LR was the focus of this study, we also looked at MCC-specific survival. We saw no difference in MCC-specific

survival between the aRT-treated and the surgery-only groups ($p=0.22$). Furthermore, within each of these groups, wide versus narrow margin size was not associated with MCC-specific survival, and results were similar when adjusted for immunosuppression, tumor size, and head/neck primary.

DISCUSSION

Previous studies have shown that wider surgical margins are associated with improved local control of MCC. However, the vast majority of these studies did not report whether or not patients also received aRT, a treatment known to be highly effective in MCC local control. This is relevant because aRT is frequently included in MCC management. Specifically, 54% of MCC patients in the National Cancer Database received aRT¹⁷. In our Seattle-based repository, among patients who had no evidence of distant metastatic disease, 92% of 826 patients received aRT to the primary site (database accessed February 2019). Whether or not aRT is included in initial management could significantly affect the appropriate surgical margin size in MCC. Indeed, the findings presented here support the concept that wider surgical margins are not indicated for patients that will receive aRT.

A recent summary of the current literature on this topic suggested that the appropriate surgical margin size for primary MCC is 1-3 cm¹⁸. However, wide margins often cause significant morbidity^{5, 19, 20}, and can delay the start of aRT if a graft or flap is required for closure⁵. The 2019 NCCN guidelines recommend “wide excision with 1- to 2-cm margins to investing fascia of muscle or pericranium when clinically feasible” but also note that if aRT is planned, then primary closure should be prioritized over wider margins.

Consistent with the existing literature, in the present cohort, among patients treated only with surgery, margin size did affect the risk of LR. Twenty percent of patients who were treated with a smaller (≤ 1 cm) surgical margin developed LR compared to 0% of patients with a larger (>1

cm) surgical margin. These findings are concordant with a study of 179 MCC patients in British Columbia¹⁴ in which surgical margin size only mattered among patients who did not receive aRT. Specifically, in the Canadian study, among patients who had narrow margin excision (<1 cm), only 5% (1/19) had LR if they received aRT, while 25% (3/12) who did not receive aRT had a LR¹⁴. Also similar to the present study, among patients who had surgical margins ≥ 1.0 cm, there was no improved local control with the addition of aRT¹⁴. In contrast, most of the existing literature does not separately consider whether or not patients received aRT. Interpretation of appropriate margin size from studies that do not describe aRT status is not feasible because of the high efficacy of aRT in controlling local disease. For example, in one of the larger studies (n=500), Fields et al. found no difference in LR rate when comparing margins <1 cm versus ≥ 1 cm but the patients were not stratified based on whether or not they received aRT⁷, making it difficult to interpret the relationship between margin size and local control.

In the present study, head and neck primary tumors were associated with a higher risk of LR compared to trunk and extremities, with 89% (8/9) of LRs occurring on the head or neck. The head/neck is a unique site where tissue-sparing surgery is important to optimize cosmetic and functional outcome, but also where radiation side effects could be morbid, especially in the elderly. In a retrospective study of 46 low risk (primary ≤ 2 cm, immunocompetent, negative SLNB, negative pathological margins) head and neck tumors from the Seattle repository, addition of aRT to the primary site significantly reduced LRs compared to surgery only¹³. Furthermore, in a separate study of 106 head/ neck MCC patients, when aRT was included, local control was over 96%²¹. In a Tampa-based, single-institution study of 113 patients with head/ neck MCC, aRT was associated with improved local control (3-year local control 89% vs 68% surgery only; $p=0.005$)²². In summary, multiple studies suggest that head and neck MCC tumors are at higher risk of recurrence following surgical monotherapy (perhaps due to limitations of margin size for this site) and that aRT should be considered for these tumors.

Given important cosmetic and functional considerations for head and neck MCC management, Mohs micrographic surgery is often considered. Currently, NCCN guidelines do not routinely recommend Mohs surgery for MCC, in part because SLNB is often indicated^{8, 23, 24}, requiring separate hospital-based surgical procedures in addition to Mohs surgery. The findings from the present study suggest that in the absence of aRT, the unique ability of Mohs surgery to attain narrow, pathologically negative surgical margins may not be as beneficial for MCC as for other skin cancers, because MCC often recurs beyond pathologically negative margins (**Table 2**; multiple patients had LR following pathologically negative excision).

Based on the results presented here and the existing literature, we have created a treatment algorithm (**Figure 3**) to aid clinicians in determining appropriate management for primary MCC tumors. Using clinical factors such as primary tumor size, primary site, and immunosuppression status, this algorithm first separates patients into a higher risk group for which aRT is indicated. Such patients can then avoid the morbidity of wide surgical margins as well as potential delays in initiating aRT. For lower risk patients who may not need aRT, clinicians may consider a wider margin with primary closure at the time of SLNB. Depending on pathology results of re-excision and SNLB, aRT may not be indicated.

Although aRT decreased LR among patients with narrower surgical margins (<1 cm), there was no difference in disease-specific survival. The findings presented here and from the literature¹⁴ demonstrate that for low risk MCCs, surgical margins >1 cm are sufficient and that aRT is not required for excellent local disease control. In contrast, if narrow surgical margins are required to reduce morbidity and obtain primary closure, there is agreement that aRT can provide excellent local control. In terms of whether survival can be affected by aRT, the present study did not observe this association. However, three cancer registry studies that were far larger than our study demonstrated that aRT was associated with significantly better overall survival^{12, 17, 25}. It is possible that with a larger sample size we might have detected survival differences based on whether or not aRT was given. Although links to survival are controversial,

current evidence suggests that optimal local control (sometimes involving aRT) can minimize LR. This is beneficial because a LR leads to patient anxiety, increased medical costs, and salvage therapies that are morbid.

Limitations of this study are its retrospective design and unavailable clinical data for some cases (50/188 patients lacked pathologic margin size and 2/188 lacked primary tumor size). Because the sample size for the surgery-only group was 48, further subgrouping of surgical margin size was not statistically feasible. Also, because our site is a tertiary referral center for MCC, patients in this cohort often received their treatment closer to their home. This cohort represents a heterogeneous group in regard to surgery and radiation therapy techniques.

This study lends support to earlier literature that suggests if localized cutaneous MCC is also treated with aRT, then narrow surgical margins are sufficient. While this study indicates that aRT plays an important role in the management of higher risk MCC tumors, it is possible that emerging approaches in aRT (e.g. a single fraction of 8 Gy radiation²⁶) may provide good local control with markedly diminished morbidity and enhanced patient convenience. As summarized in the flowchart (**Figure 3**), we believe that surgical margins for a patient's MCC should be determined with careful consideration of risk factors and the potential role of aRT in optimizing a patient's outcome.

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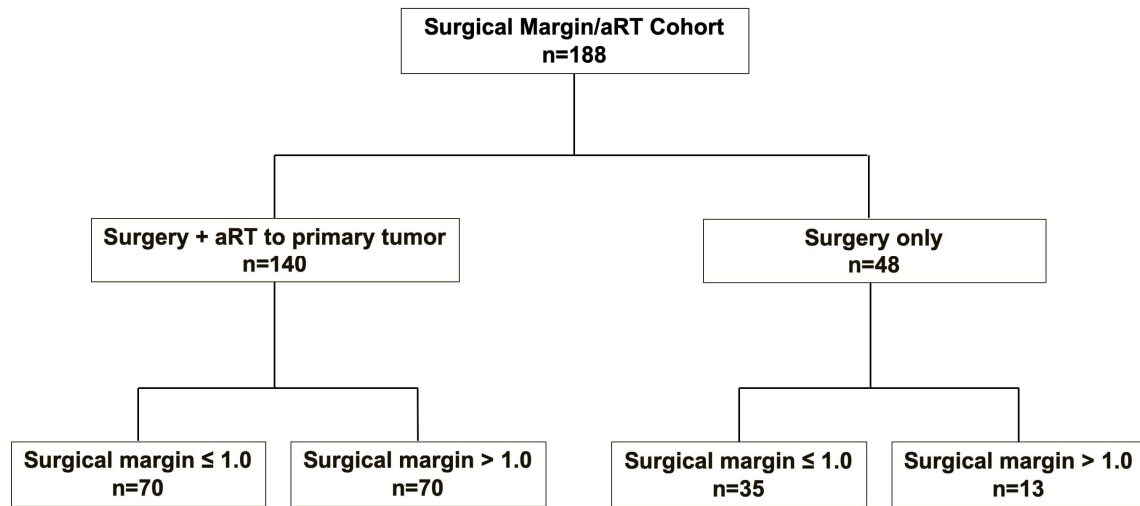


Figure 1: Flowchart of MCC patients included in this surgical margin/aRT cohort. All 188 patients met the following 5 entry criteria: local-only MCC by clinical exam at diagnosis (stage pathological-I, clinical-I, pathological-IIA, clinical-IIA, pathological-IIIA), underwent primary lesion excision, aRT status available, surgical margin status available, and enrolled within 180 days from diagnosis.

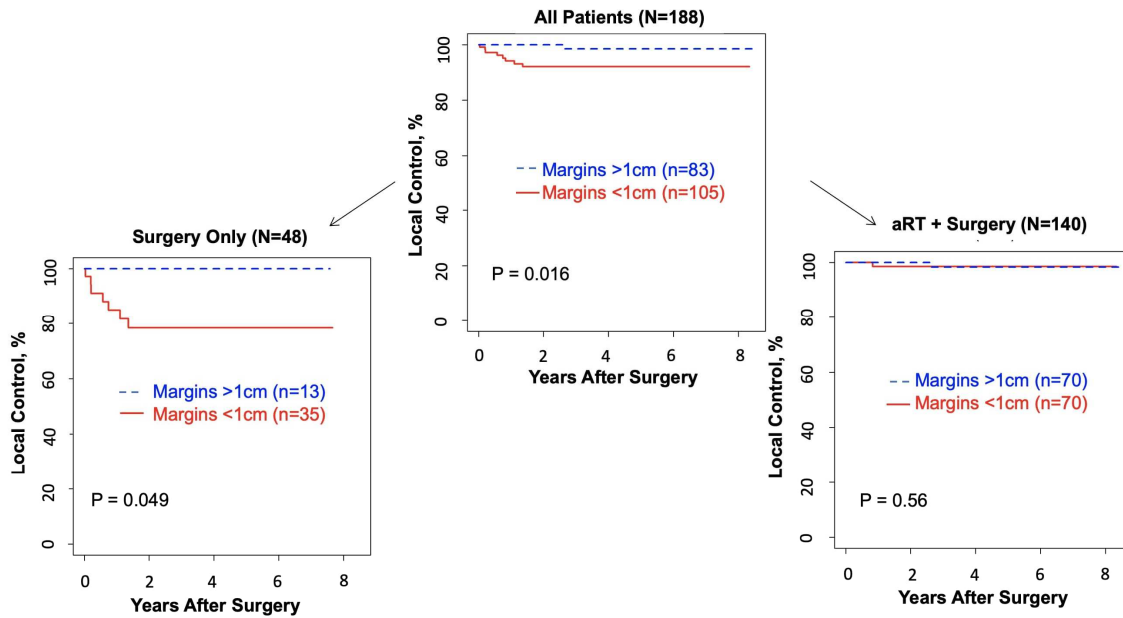


Figure 2: Control of Merkel cell carcinoma at the primary site as a function of surgical margin size and whether or not adjuvant radiation was given. Cumulative incidence curves are shown, with death and any non-local MCC recurrence being competing risks.

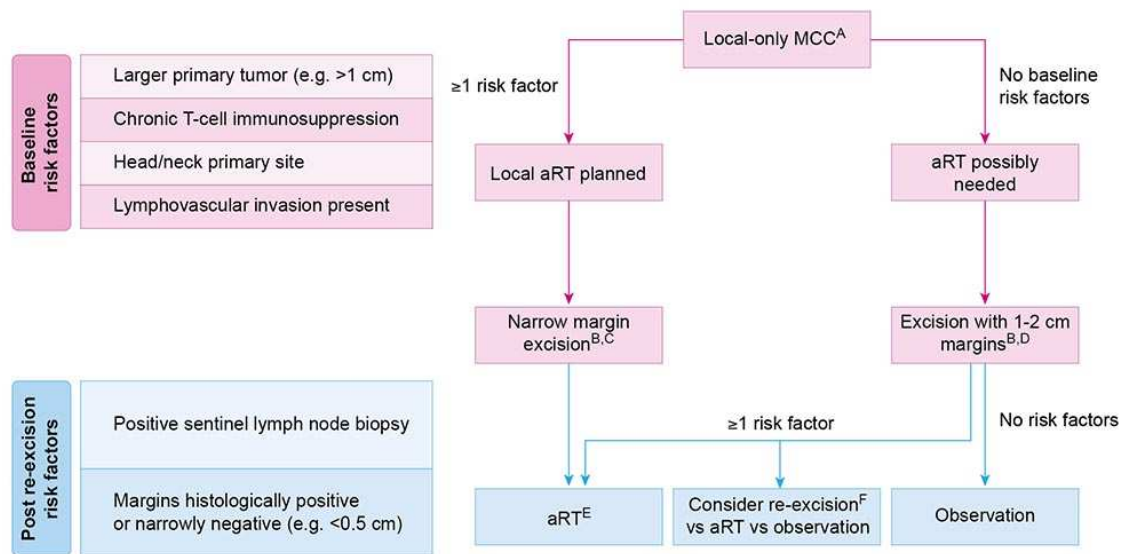


Figure 3. Suggested local MCC treatment management. This flowchart that integrates treatment options with risk factors that affect are associated with local recurrence. Certain risk factors are available at time of diagnosis (baseline), while others are only available after surgical excision (post re-excision).

^A Criteria for “local only MCC” were clinically node-negative, no in-transit disease, and imaging negative for distant disease.

^B Sentinel lymph node biopsy (SLNB) typically performed at this time.

^C Narrow excision margins minimize morbidity and if aRT is performed microscopically positive margins are acceptable.

^D Goal should be primary tissue closure (i.e. without a flap or graft) allowing aRT initiation within 3-4 weeks^{5, 19, 20}.

^E If SLNB positive, nodal aRT would typically be given, in addition to primary site aRT.

^F Decision on re-excision based on clinical setting (narrow path margins, e.g. <0.5 cm) and patient preference, re-excision vs aRT vs observation.

Table 1: Comparison of clinical and tumor characteristics between all patients and specified subgroups.

Variable	All Patients (n=188)	Surgery + aRT		P- value*	Surgery Only		P-value*
		Margins ≤1 cm (n=70)	Margins >1 cm (n=70)		Margins ≤1 cm (n=35)	Margins >1 cm (n=13)	
Female sex	72 (38.3)	26 (37.1)	22 (31.4)	0.59	16 (45.7)	8 (61.5)	0.52
Age ≥ 65	121 (64.4)	44 (62.9)	46 (65.7)	0.86	24 (68.6)	7 (53.8)	0.50
Immune suppressed	18 (9.6)	10 (14.3)	5 (7.1)	0.27	3 (8.6)	0 (0.0)	0.55
Nodal stage (path stage IIIA)**	45 (23.9)	17 (24.3)	25 (35.7)	0.20	2 (5.7)	1 (7.7)	>0.99
Head & neck primary***	66 (35.1)	26 (37.1)	10 (14.3)	0.003	25 (71.4)	5 (38.5)	0.049
Size of primary †				0.020			0.92
≤1 cm	82 (44.1)	30 (42.9)	18 (26.5)		25 (71.4)	9 (69.2)	>0.99
1-2 cm	60 (32.3)	25 (35.7)	25 (36.8)		7 (20.0)	3 (23.1)	
≥ 2 cm	44 (23.7)	15 (21.4)	25 (36.8)		3 (8.6)	1 (7.7)	
SLNB performed	164 (87.2)	63 (90.0)	65 (92.9)	0.76	25 (71.4)	11 (84.6)	0.47
Unknown	36 (19.1)	13 (18.6)	17 (24.3)		3 (8.6)	3 (23.1)	

*Fisher's exact test or the Wilcoxon rank-sum test (size of primary)

**Compared to local stage of diagnosis

***Compared to primaries on extremities and trunk

†Two patients did not have a tumor size available

Table 2: Characteristics of 9 patients who experienced local recurrence of MCC.

Patient ID**	aRT	Age/ Gender	Stage*	Site of primary	Size of Primary (cm)	Surgical Margin (cm)	LVI	Narrowest radial margin via pathology (cm)
1	No	73 M	C-I	Head & Neck	0.7	0.5	Unknown	No residual tumor
2		59 F	P-I	Head & Neck	0.5	1.0	Unknown	0.2
3		76 M	P-I	Head & Neck	0.5	1.0	Absent	No residual tumor
4		58 M	P-I	Head & Neck	0.5	1.0	Absent	No residual tumor
5		84 M	P-I	Head & Neck	0.4	1.0	Absent	No residual tumor
6		80 M	P-I	Head & Neck	0.8	1.0	Present	0.4
7		70 F	P-I	Head & Neck	0.6	1.0	Absent	0.4
8	Yes	71 F	P-IIA	Lower Limb	2.4	2.0	Unknown	Unknown
9		67 M	P-I	Head & Neck	0.8	1.0	Absent	Unknown

*Stage is according to AJCC 8th edition.

** No patients were immunosuppressed

